Module 4 – Scalable Architecture

1. Folder Structure

What is Folder Structure?

Folder structure refers to how files and directories are organized in a React project.

Why It Matters

• Improves code readability
• Makes project easy to maintain
• Helps teams collaborate efficiently

Example Structure:

```
src/
 ├── components/
 ├── pages/
 ├── hooks/
 ├── context/
 ├── services/
 ├── utils/
 └── App.js
```

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2. Reusable Components

What are Reusable Components?

Reusable components are components that can be used multiple times across the application.

Benefits

• Reduces code duplication
• Improves consistency
• Saves development time

Example:

```
function Button({ text }) {
  return <button>{text}</button>;
}
```

Usage:

```
<Button text="Login" />
<Button text="Signup" />
```

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3. State Management

What is State Management?

State management is the process of handling and sharing data across components.

Types of State

• Local State (inside component)
• Global State (shared across app)

Key Points

• Use local state for small components
• Use global state for large apps
• Avoid unnecessary prop drilling

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4. Redux Basics

What is Redux?

Redux is a state management library used to manage global state in large applications.

Core Concepts

• Store – Holds application state
• Actions – Describe what to do
• Reducer – Updates state

Example:

```
const initialState = { count: 0 };

function reducer(state = initialState, action) {
  switch (action.type) {
    case "INCREMENT":
      return { count: state.count + 1 };
    default:
      return state;
  }
}
```

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5. Best Practices

Important Practices

• Keep components small and focused
• Use meaningful file and variable names
• Avoid deeply nested components
• Separate logic and UI
• Use consistent folder structure

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6. Key Concepts to Remember

• Organized structure improves scalability
• Reusable components save time
• Proper state management is crucial
• Redux helps manage large-scale state
• Following best practices ensures clean code

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7. Summary

Scalable architecture is essential for building large and maintainable React applications. By organizing your project properly, creating reusable components, managing state efficiently, and following best practices, you can build applications that are easy to scale and maintain.